

Quiz 8

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (5 points) Test the following series for convergence or divergence.

(a) $\sum_{n=1}^{+\infty} \frac{3^n}{n!}$ (Using the Ratio Test)

$$\begin{aligned} a_n &= \frac{3^n}{n!} \\ \lim_{n \rightarrow +\infty} \left| \frac{a_{n+1}}{a_n} \right| &= \lim_{n \rightarrow +\infty} \left| \frac{\frac{3^{n+1}}{(n+1)!}}{\frac{3^n}{n!}} \right| \\ &= \lim_{n \rightarrow +\infty} \frac{3^{n+1}}{(n+1)!} \cdot \frac{n!}{3^n} \\ &= \lim_{n \rightarrow +\infty} \frac{3}{n+1} = 0 < 1 \\ \Rightarrow \sum_{n=1}^{+\infty} \frac{3^n}{n!} &\text{ converges by the} \\ &\text{Ratio Test} \end{aligned}$$

(b) $\sum_{n=1}^{+\infty} \frac{n}{2^n}$ (Using the Root Test)

$$\begin{aligned} a_n &= \frac{n}{2^n} \\ \lim_{n \rightarrow +\infty} \sqrt[n]{|a_n|} &= \lim_{n \rightarrow +\infty} \left(\frac{n}{2^n} \right)^{\frac{1}{n}} \\ &= \lim_{n \rightarrow +\infty} \frac{\sqrt[n]{n}}{2} \\ &= \frac{1}{2} < 1 \\ \Rightarrow \sum_{n=1}^{+\infty} \frac{n}{2^n} &\text{ converges by} \\ &\text{the Root Test} \end{aligned}$$

Problem 2. (5 points) Test the following alternating series for conditional convergence, absolute convergence or divergence.

(a) $\sum_{n=1}^{+\infty} \frac{(-1)^{n+1}}{2n-1}$

$$\begin{aligned} u_n &= \left| \frac{(-1)^{n+1}}{2n-1} \right| = \frac{1}{2n-1} \\ \text{i) It is eq to see that} \\ u_{n+1} &= \frac{1}{2(n+1)-1} < \frac{1}{2n-1} = u_n \text{ for } n \geq 1 \\ \text{ii) } \lim_{n \rightarrow +\infty} u_n &= \lim_{n \rightarrow +\infty} \frac{1}{2n-1} = 0 \\ \Rightarrow \sum_{n=1}^{+\infty} \frac{(-1)^{n+1}}{2n-1} &\text{ converges by the} \\ &\text{Alternating Series Test.} \end{aligned}$$

$$\begin{aligned} \text{Also } u_n &= \frac{1}{2n-1} > \frac{1}{2n} \text{ for } n \geq 1 \\ \text{and } \sum_{n=1}^{+\infty} \frac{1}{2n} &\text{ diverges} \\ \Rightarrow \sum_{n=1}^{+\infty} \frac{1}{2n-1} &\text{ diverges} \end{aligned}$$

$$\text{Thus } \sum_{n=1}^{+\infty} \frac{(-1)^{n+1}}{2n-1} \text{ is conditionally convergent.}$$

(b) $\sum_{n=1}^{+\infty} (-1)^{n+1} \frac{n}{3n+5}$

$$\begin{aligned} u_n &= \frac{n}{3n+5} \\ \lim_{n \rightarrow +\infty} u_n &= \lim_{n \rightarrow +\infty} \frac{n}{3n+5} \\ &= \lim_{n \rightarrow +\infty} \frac{1}{3 + \frac{5}{n}} = \frac{1}{3} \neq 0 \\ \Rightarrow \sum_{n=1}^{+\infty} (-1)^{n+1} \frac{n}{3n+5} &\text{ diverges} \end{aligned}$$